

CS474 HW 4 Report

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Problem Overview:

In this week's installment of assignments, I utilized the sci-kit learn package's SVC classification system to differentiate between handwritten digits numbered 1 to 5. For the data, there were images that were instrumented through a 16x16 matrix of grayscale pixel values from -1 to 1. Each digit image was represented using two extracted features that were symmetry and average intensity. The classification was performed in a 2-dimension feature space. This report documents the effect of the different values of C (cost parameter) using a linear kernel and other different kernel types.

Different C-Values:

C-Value	Accuracy	Total Support Vectors
0.01	0.8939	1080
0.1	0.9623	414
1	0.9599	162
2	0.9599	129
3	0.9623	117
5	0.9623	104

Figure 1: Part A results

The results in Figure 1 show how the miniscule values of C led to the worst performance. When $C = 0.01$, the accuracy was the lowest by far and had the most support vectors, which indicates that the classifier was weak and allowed too many misclassified samples over the decision boundary. When the C value was increased to 0.1 and above, the accuracy stagnated at around 96%. The number of support vectors jumped to 414 at 0.1, and then remained constantly at around 100 with C values 1 and above. This support vector amount decreased lightly as the c value increased from 1 and above. This means that fewer training points were needed to define the decision boundary for bigger values of C.

Different Kernel Values:

Kernel Type	Accuracy	Total Support Vectors
Linear	0.9599	162
Polynomial	0.9575	75
Rbf	0.9623	90

Figure 2: Part B results

In Figure 2, linear, polynomial, and RBF kernels are tested for their performance regarding testing accuracy and support vectors. The results indicate that the RBF kernel achieved the

highest accuracy by a slight margin, while the polynomial kernel used the least amount of total support vectors. The linear kernel performed the worst overall, with nearly double the amount of total support kernels and yet a pretty similar accuracy rating. These results show that the 2-dimensional feature space is fairly separable, but a nonlinear decision boundary provides the best performance. It also shows that the different kernels have extremely similar testing accuracy despite having differing amounts of total support kernels, which surprises me immensely.

Conclusion:

The best performing models for each of the assignment parts were the linear SVM with moderately large values of C and the RBF kernel, which both had around 96% testing accuracy on recognizing the handwritten digits in the dataset. From the part A experiment, I learned that smaller C values can damage performance and greatly increase the support vector amount, while larger values lead to more stable results and a simpler boundary. From the kernel comparison in part B, I learned that the data is close to being linearly separable, but allowing nonlinearity through the RBF kernel has the best testing-set performance. I also learned that different kernels can have differing amounts of total support vectors and yet perform similarly in accuracy.